State Machine Formulation

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State Machine

$$M = (S, s_0, E, \Sigma, T)$$

- S: state set
- s_0: initial state
- E: event set
- Σ: data context
- T: transitions

Transitions Set

$$T \subseteq S \times E \times G \times A \times S$$

- T: transition set
- S: state set
- E: event set
- G: guard functions

Transition Instance

$$t = (s_i, \tau, g, a, s_{i+1})$$

- t: transition
- $s_i \in S$: source state
- $\tau \in E$: trigger event
- $g \in G$: guard function
- $a \in A$: guard functions

Guard and Action Function

$$g : \Sigma \to \mathbb{B}$$

$$a : \Sigma \to \Sigma \times \Omega$$

- $\mathbb{B} = \{\mathit{false}, \mathit{true}\}$
- Ω: set of side effects

Complex state machine

$$M_C = (C, v_0, E, \Sigma, H)$$

- C: current configuration
- v_0: initial vertex
- E: event set
- Σ: data context
- H: hyper-transitions

Hyper transition

$$H = (\rho, \tau, \phi, \alpha)$$

- H: hyper-transition
- ρ: hyper edge
- $\tau \in E$: trigger event
- ϕ: reduction function
- α: complex action

Selector Function

$$\phi : \mathcal{P}(V) \times \mathcal{P}(V) \to \mathcal{P}(V) \times \mathcal{P}(V)$$

$$(V_S, V_T) \to (\overline{V}_S, \overline{V}_T), \overline{V}_S \subseteq V_S \wedge \overline{V}_T \subseteq V_T$$

$$v_s \in V_s \wedge \overline{V}_T \subseteq V_T$$

Transition Activity Function

$$\alpha : \overline{\mathrm{P}} \times E \times \Sigma \times C \rightarrow \Sigma \times C \times \Omega$$

$$\overline{\mathrm{P}} = \{\overline{\rho} \mid \overline{\rho} = (\overline{V}_S, \overline{V}_T) \wedge \overline{V}_S, \overline{V}_T \in \mathcal{P}(V)\}$$

Parallel Execution Context

$$\widetilde{\alpha} = \phi \times \Lambda$$